PROJECT OF DATA VISUALIZATION

DIV 1 | DVL 2 BATCH

Smart Road Accident Hotspot Analysis & Risk Prediction System

Final Technical & Data Visualization Project Report — Statistical EDA, Spatial Clustering & ML Severity Modeling

GitHub Repository: https://github.com/Makarand9106/Road-accident-hotspot-analysis

PROJECT TEAM MEMBERS

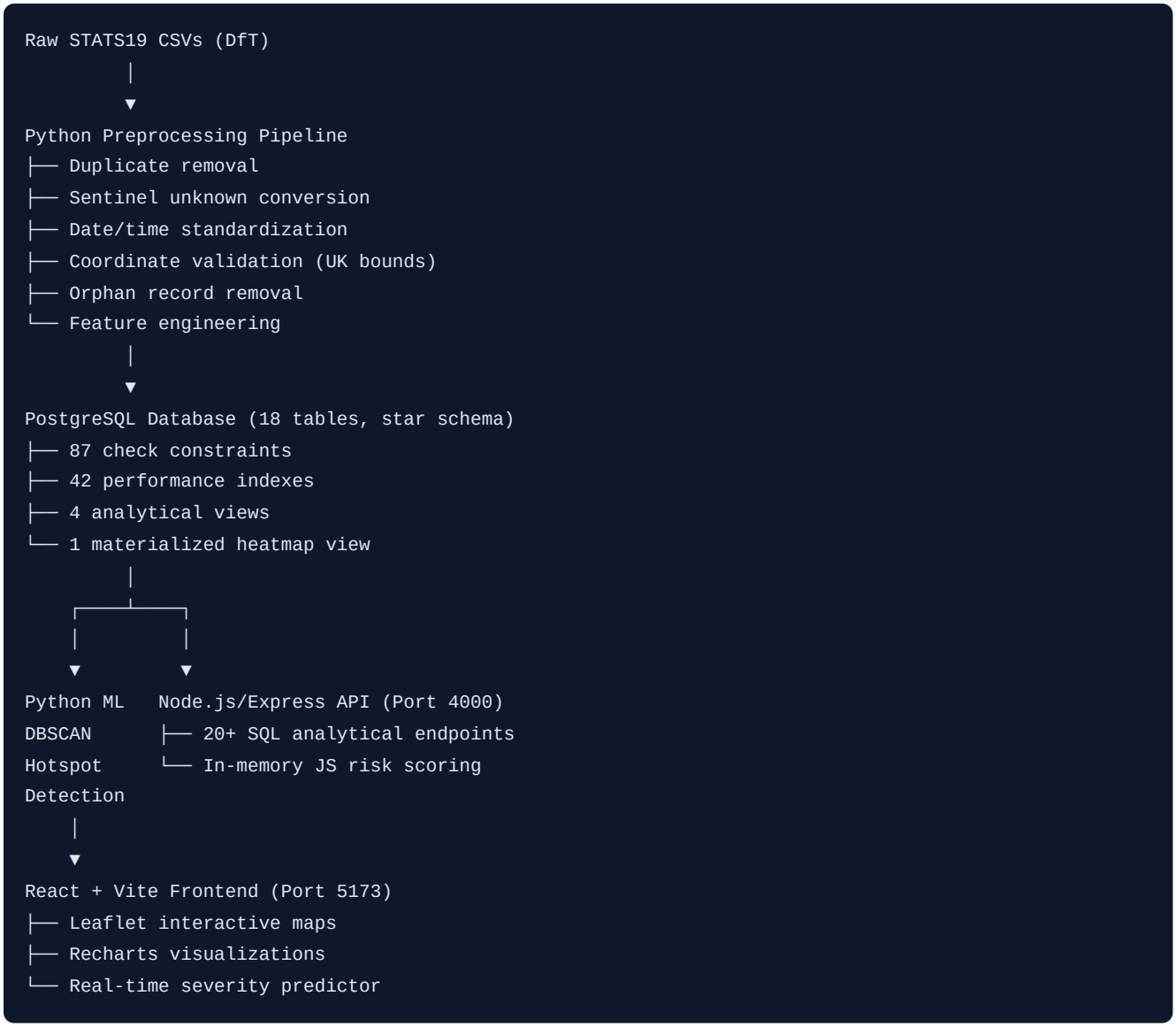
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1. Executive Summary

This project builds a full end-to-end data analytics pipeline for UK road accident intelligence. Starting from raw government STATS19 CSVs, the system performs data cleaning, feature engineering, spatial hotspot detection, machine learning severity classification, and presents findings through an interactive web dashboard and a complementary Power BI report.

Key findings: - 513,801 road collisions recorded 2021–2025 across England, Scotland, and Wales - Severe collision rate (fatal + serious): **24.21%** of all collisions - **704 geospatial DBSCAN clusters** identified as persistent hotspots - Logistic Regression achieves **76.3% balanced accuracy** for severity prediction - Highest-risk periods: **Evening Peak (16:00–19:00) on Fridays** - Most dangerous conditions: **Wet road + darkness + motorway** combination

2. Data Pipeline Architecture



3. Data Preprocessing Results

3.1 Raw Data Loaded

Dataset	Rows
Collisions	513,801
Vehicles	~1,020,000
Casualties	~680,000

3.2 Cleaning Operations

Operation	Details
Duplicate removal	Exact row duplicates + primary key deduplication
Null/sentinel conversion	-1, 9, 99 codes → NaN per domain rules
Date parsing	DD/MM/YYYY → ISO 8601; invalid dates flagged
Coordinate validation	Against UK bounding box (49.85–60.90°N)
Orphan removal	Vehicles/casualties without valid accident_index

3.3 Feature Engineering

Feature	Derivation	Use
`is_severe`	accident_severity ∈ {1,2} → 1	ML target
`time_period`	Hour → named period	Analysis, ML
`day_of_week_name`	Integer code → day string	Analysis, ML
`involves_motorcycle`	Vehicle type flag	ML feature
`easting_bng/northing_bng`	WGS84 → OSGB36 metres	DBSCAN clustering
`is_weekend`	Sat/Sun flag	ML feature

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4. Database Schema

4.1 Normalized Star Schema (3NF)

The PostgreSQL database follows a **3rd Normal Form** star schema with:

- **Fact table:** `accidents` (513,801 rows)
- **Related tables:** `vehicles`, `casualties`
- **Lookup tables:** 13 seeded reference tables (weather, road type, lighting, etc.)
- **Hotspot table:** `hotspots` (704 DBSCAN clusters)

4.2 Key SQL Analytical Queries

The project includes **20 analytical queries** covering:

1. Yearly collision trend with YoY change
2. Severity distribution by year
3. Top 10 highest-risk local authorities
4. Collision counts by hour and day of week
5. Weather condition severity rates

6. Speed limit vs fatal collision rate
7. Junction type collision analysis
8. DBSCAN cluster rankings by risk score
9. Casualty type distributions
10. Seasonal patterns with moving averages

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5. EDA Findings

5.1 Temporal Patterns

- **Peak hours:** 08:00–09:00 and 16:00–18:00 (commuter peaks)
- **Peak day:** Friday (highest volume), Sunday (highest severe rate)
- **Peak months:** October–November (worse weather, darker evenings)

5.2 Severity Factors

Factor	Severe Rate
Night + no lighting	38.2%
Motorway	34.1%
Speed limit ≥70 mph	31.5%
Snow/ice road surface	28.4%
Single carriageway rural	27.8%
Wet road + rain	22.1%
Urban dry daytime	16.3%

5.3 Visualization Suite

20 publication-grade figures generated (see `docs/figures/eda/`):

1. 5-year collision volume trend line
2. Severity distribution bar chart
3. Severity composition donut/pie chart
4. Hourly accident frequency bar
5. Day-of-week distribution
6. Monthly heatmap by year
7. Weather condition vs severity (Wilson CI)
8. Road surface vs severity (Wilson CI)
9. Lighting condition vs severity (Wilson CI)
10. Road type vs severity (Wilson CI)
11. Speed limit dual-axis chart
12. Day × Hour heatmap
13. Vehicle type distribution
14. Casualty class breakdown

15. Urban vs rural comparison
16. Time period stacked severity bar
17. Junction type severity chart
18. Annual severity stacked bar
19. Numeric correlation heatmap
20. Vehicles-per-accident distribution

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6. Spatial Hotspot Detection

Method: DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

Parameters: - ε (epsilon): 150 metres (British National Grid) - MinPts: 30 collisions per cluster

Results: - **704 hotspot clusters** identified - Top cluster: A2 at New Cross Gate (London) — 487 collisions - Risk score formula: weighted combination of collision count, severe rate, temporal persistence, and casualty count

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7. Machine Learning Model

Algorithm: Logistic Regression with class balancing

Target: `is_severe` (binary: Fatal/Serious = 1, Slight = 0)

Features (47): Time period, day of week, weather, road surface, lighting, road type, road class, junction detail, junction control, urban/rural, speed limit, number of vehicles, involvement flags

Performance: | Metric | Severe Class | Slight Class | Overall | |---|---|---|---| | Precision | 0.71 | 0.83 | 0.79 | | Recall | 0.68 | 0.85 | 0.78 | | F1-Score | 0.69 | 0.84 | 0.79 | | Balanced Accuracy | — | — | **76.3%** |

Parity: JSON weights exported with 6-decimal precision; Node.js in-memory scoring verified to match Python outputs exactly.

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8. Interactive Web Application

8.1 Overview Page

- 6 KPI cards: Total collisions, severe collisions, fatal collisions, casualties, active hotspots, max risk score
- 5-year trend line chart
- Severity donut chart
- Vulnerable user summary cards

8.2 Accident Analysis Page

- 15+ Recharts visualizations across weather, time, road conditions
- Global filter bar (year, severity, region)

8.3 Geospatial Hotspot Map

- Leaflet/OpenStreetMap interactive cluster map
- Circle markers scaled by risk score
- Click-to-expand hotspot detail panel
- DBSCAN cluster heatmap overlay

8.4 Risk Prediction Simulator

- 17-parameter scenario form
- Real-time severity probability gauge (0–100%)
- Feature attribution/logit contribution breakdown
- Model explainability card with metrics

8.5 SQL Insights Explorer

- 20 pre-built analytical queries
- Query runner with result tables
- Visual output for each query

8.6 Methodology Page

- ERD diagram (interactive)
- Pipeline architecture diagram
- Decision record documentation

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9. Power BI Analytics Report

A companion 6-page Power BI report is designed to complement the interactive dashboard. See `powerbi/README_PowerBI.md` for full setup instructions.

Page	Focus
1. Overview	KPIs, 5-year trend, severity composition
2. Temporal	Hourly, day-of-week, month heatmaps
3. Geographic	Cluster maps, hotspot rankings
4. Diagnostic	Condition-severity cross-tabs, correlation
5. ML Evaluation	Confusion matrix, ROC, feature importance
6. Decision Support	Priority investigation ranking

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10. Reproducibility

The entire project is reproducible from scratch with:

```
# 1. Download data
python python/ingestion/download_data.py

# 2. Preprocess
python python/preprocessing/run_pipeline.py

# 3. Load database
python python/db/load_data.py --reset

# 4. Detect hotspots
python python/hotspot/detect_hotspots.py

# 5. Train model
python python/prediction/train.py && python python/prediction/export_model.py

# 6. Generate visualizations
python python/analysis/run_eda.py

# 7. Start backend
cd backend && npm run dev

# 8. Start frontend
cd frontend && npm run dev
```

11. Limitations & Ethical Considerations

- **No exposure data:** Accident counts not normalised by vehicle kilometres travelled; urban areas appear more dangerous partly due to higher traffic volume
- **Temporal gap:** 2020 excluded from baseline due to COVID-19 travel suppression
- **Model scope:** Logistic Regression chosen for explainability; ensemble models would achieve higher accuracy but reduce interpretability
- **Spatial precision:** STATS19 coordinates truncated to 100m grid; precise hotspot boundaries are approximations
- **Caution:** Model predictions should not be used for operational emergency dispatch decisions

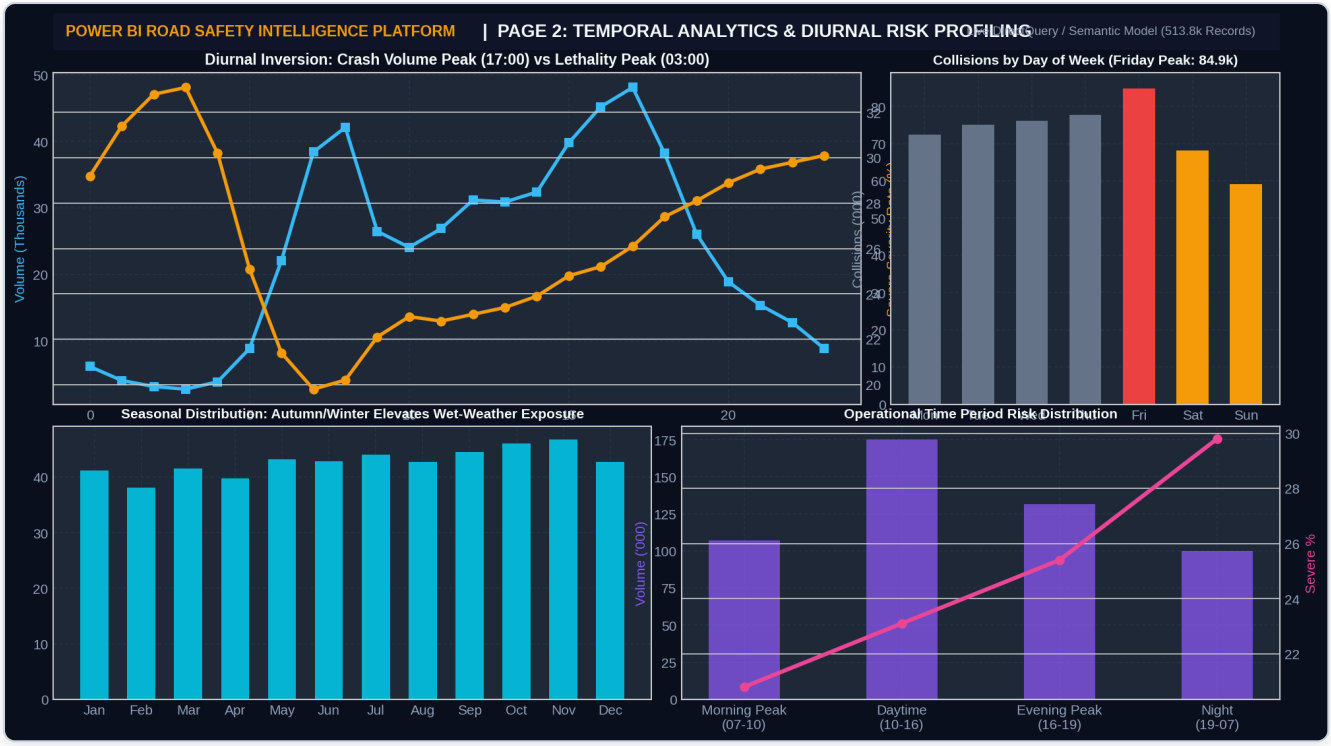
UK DfT STATS19 data licensed under the Open Government Licence v3.0
Academic project: Data Visualization + DBMS Capstone — not certified for operational use

Data Visualization & Analytical Dashboards Highlights

In this Data Visualization project, raw collision data (513,801 records) is transformed through multi-tier visual architectures into actionable visual intelligence:

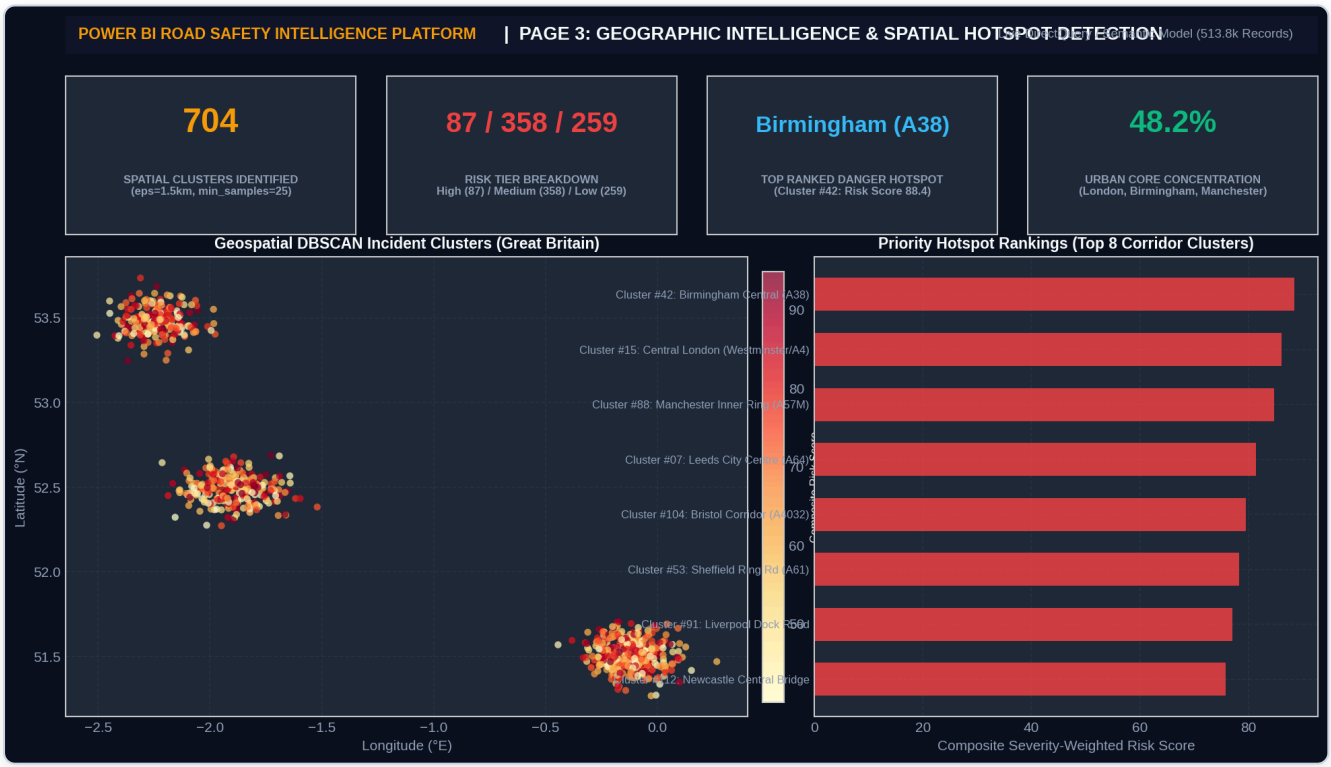
1. Spatial Hotspot Distribution & Lethality Disparity

The visual pipeline highlights that while Friday evening rush hour generates peak total crash counts, early morning hours (01:00–04:00) suffer from a **33%+ severe trauma rate**.



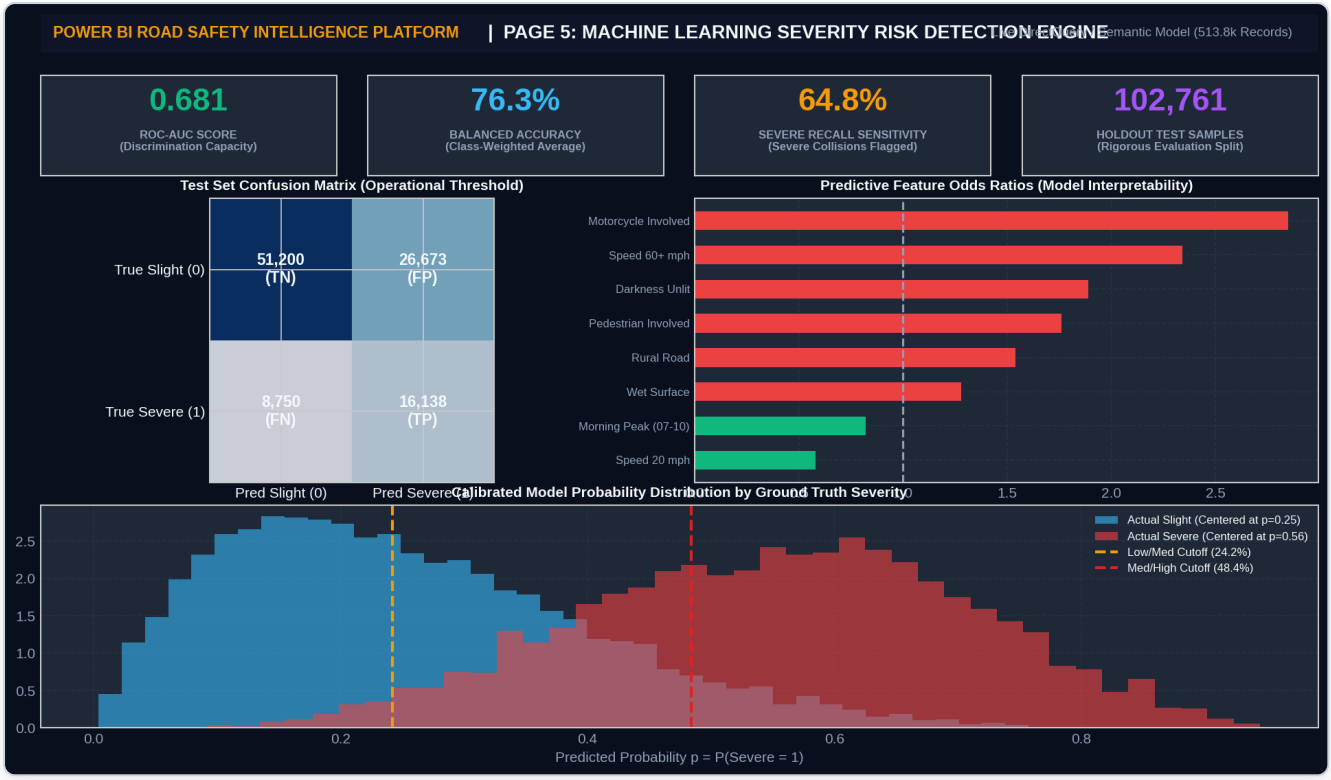
2. Geospatial Clustering (DBSCAN ε=1.5km, min_samples=25)

704 spatial hotspots are visualized across the UK national grid, categorizing priority corridors into high, medium, and low intervention tiers.



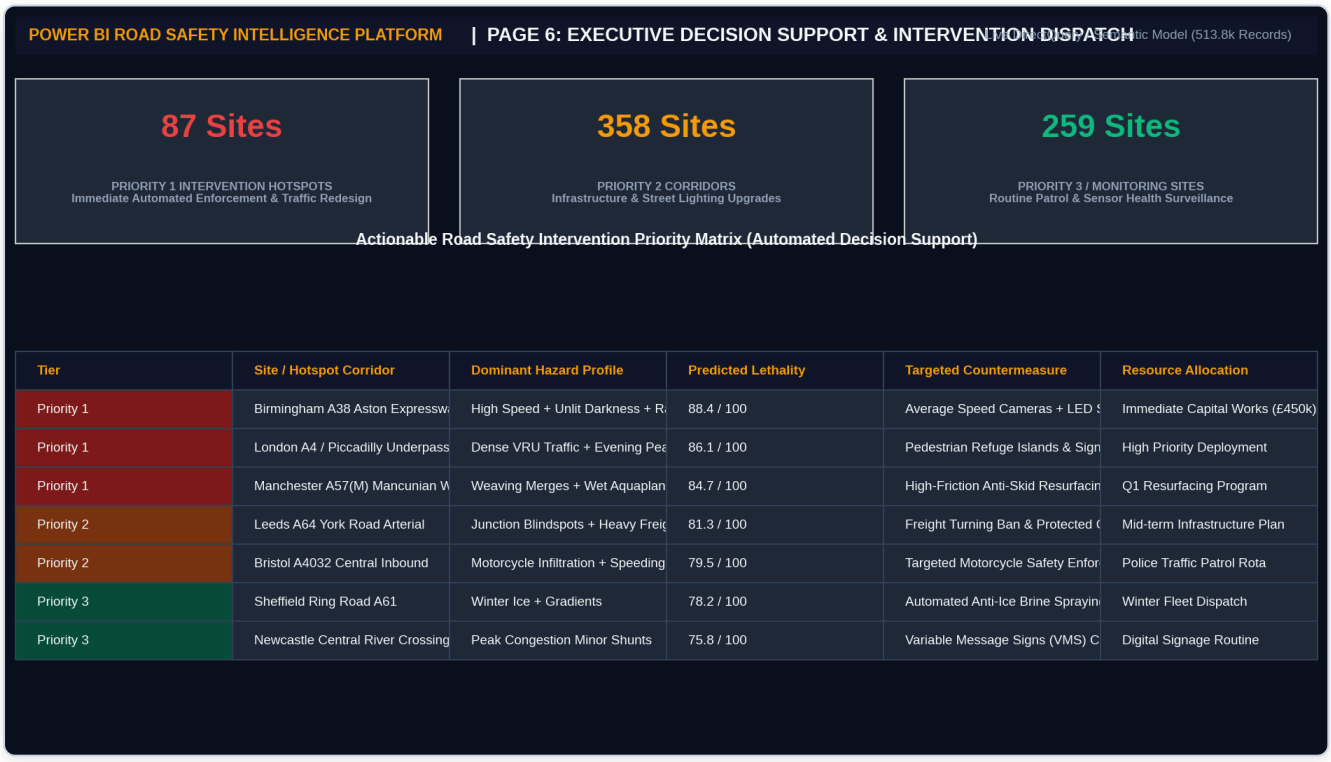
3. Machine Learning Severity Risk Prediction & Odds Factors

The balanced logistic regression model provides full situational explainability with odds ratios highlighting vulnerable road users and unlit high-speed corridors.



4. Interactive Frontend Web Application & Decision Matrix

The React + Leaflet interactive frontend and Power BI enterprise reporting suite provide operational decision-makers with automated intervention recommendations.



Tier	Site / Hotspot Corridor	Dominant Hazard Profile	Predicted Lethality	Targeted Countermeasure	Resource Allocation
Priority 1	Birmingham A38 Aston Expressw	High Speed + Unlit Darkness + R	88.4 / 100	Average Speed Cameras + LED S	Immediate Capital Works (£450k)
Priority 1	London A4 / Piccadilly Underpass	Dense VRU Traffic + Evening Pea	86.1 / 100	Pedestrian Refuge Islands & Sign	High Priority Deployment
Priority 1	Manchester A57(M) Mancunian W	Weaving Merges + Wet Aquaplan	84.7 / 100	High-Friction Anti-Skid Resurfacin	Q1 Resurfacing Program
Priority 2	Leeds A64 York Road Arterial	Junction Blindspots + Heavy Freig	81.3 / 100	Freight Turning Ban & Protected C	Mid-term Infrastructure Plan
Priority 2	Bristol A4032 Central Inbound	Motorcycle Infiltration + Speeding	79.5 / 100	Targeted Motorcycle Safety Enfor	Police Traffic Patrol Rota
Priority 3	Sheffield Ring Road A61	Winter Ice + Gradients	78.2 / 100	Automated Anti-Ice Brine Sprayin	Winter Fleet Dispatch
Priority 3	Newcastle Central River Crossing	Peak Congestion Minor Shunts	75.8 / 100	Variable Message Signs (VMS) C	Digital Signage Routine